

ARTHASHASTRA 2.0

The Analytical Companion

Thirty Sutras · Governance Scores · Indicators · Actions · Risks

Md Taufiqul Islam

Joint Secretary, Cabinet Division, Government of Bangladesh

ANALYTICAL DIMENSIONS

- | | |
|----------------------|--|
| ● URGENCY | How soon action is required |
| ● FEASIBILITY | How achievable with current capacity |
| ● IMPACT | Magnitude of consequence if acted upon |

Methodology: Reading the Analytical Panels

Each of the thirty Sutras is accompanied by a structured analytical panel designed for direct policy use. Three governance scores are assigned on a ten-point scale: **Urgency** captures how rapidly the window for effective action is closing; **Feasibility** assesses achievability given Bangladesh's present institutional, fiscal, and technical capacity; **Impact** estimates the magnitude of national consequence if the Sutra is acted upon — or ignored. Scores are the author's structured judgments, calibrated against the Oxford Insights Government AI Readiness Index 2025, the UNESCO AI Readiness Assessment for Bangladesh (2025), and the scholarly literature anchored in each panel.

Each panel further specifies **Key Indicators** — measurable quantities through which progress can be tracked in annual reporting; **Priority Actions** — concrete first moves implementable within existing constitutional and administrative structures; a **Risk-if-Ignored** statement articulating the failure mode the Sutra guards against; and a **Scholarly Anchor** connecting the maxim to the peer-reviewed literature. Panels are colour-coded by theme: Governance (navy), Economy (teal), Security (rust), Ethics (gold), and Resilience (blue).

The companion closes with a thirty-row summary heatmap enabling rapid triage: Sutras scoring high on both Urgency and Impact but low on Feasibility identify where capacity-building investment is the binding constraint; high-Feasibility, high-Impact Sutras identify the quick wins with which implementation should begin.

SUTRA 1 · CHAPTER I

GOVERNANCE

The ruler shall listen to both the voices of his people and machine judgment, yet surrender to neither.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- % of cabinet-level decisions with documented human-override authority
- Existence of an AI Decision-Audit Register in each ministry
- Annual survey: citizen trust in AI-assisted public decisions

PRIORITY ACTIONS

- Mandate human-in-the-loop sign-off for all consequential automated decisions
- Train Deputy Secretaries and above in algorithmic literacy
- Publish an annual AI Decision Transparency Report

Risk if ignored: Decision-making authority quietly migrates to opaque systems and the vendors who build them; accountability dissolves.

Scholarly anchor: Pasquale (2015), *The Black Box Society*, Harvard University Press

SUTRA 2 · CHAPTER I

GOVERNANCE

He who does not understand code shall be ruled by those who write it.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- % of senior civil servants completing certified AI literacy training
- Number of technologists embedded in policy units
- Procurement contracts independently auditable by government engineers

PRIORITY ACTIONS

- Compulsory AI fundamentals module at BPATC for all cadres
- Create a Government Technologist cadre with competitive pay
- Require source-code escrow for all critical government AI systems

Risk if ignored: Vendor capture: the state becomes structurally incapable of evaluating what it buys, deploys, or regulates.

Scholarly anchor: Acemoglu (2021), *Harms of AI*, NBER WP 29247

SUTRA 3 · CHAPTER I

GOVERNANCE

In a land of many rivers, governance must flow — adaptive, not rigid.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Statutory review interval of the National AI Policy (target: annual)
- Number of regulatory sandbox cohorts completed per year
- Average time from policy gap detection to amendment

PRIORITY ACTIONS

- Insert sunset-and-review clauses into every AI regulation
- Operationalise National AI Sandboxes with published learning reports
- Establish a horizon-scanning cell in the Cabinet Division

Risk if ignored: Static rules ossify; regulation lags technology by years and either strangles innovation or fails to prevent harm.

Scholarly anchor: Walker et al. (2004), *Ecology & Society* 9(2)

SUTRA 4 · CHAPTER II

ECONOMY

Data is the new wealth; guard it as once gold was guarded in vaults.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- % of government data hosted on sovereign infrastructure
- Enactment status of the Data Protection Ordinance
- Volume of open Bangla-language datasets published (GB/year)

PRIORITY ACTIONS

- Enact the Data Protection Ordinance with an independent authority
- Build the National Data Trust as a distributed edge-node grid
- Mandate sovereign hosting for all citizen-service data

Risk if ignored: National data wealth accrues to foreign clouds; Bangladesh becomes a data colony with no bargaining position.

Scholarly anchor: Acquisti, Taylor & Wagman (2016), J. of Economic Literature 54(2)

SUTRA 5 · CHAPTER II

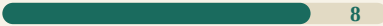
ECONOMY

Garments are woven in factories, but power is woven in networks.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- AI/data services export earnings (USD M/year)
- Number of certified ethical data-annotation firms
- Average wage premium of data workers vs. RMG baseline

PRIORITY ACTIONS

- Launch an Ethical Data Annotation export strategy with fair-labour certification
- Extend RMG-style bonded-zone incentives to AI services parks
- Negotiate data-services chapters into trade agreements

Risk if ignored: The garment playbook is never translated; the demographic dividend expires unconverted into digital value.

Scholarly anchor: Sambasivan et al. (2021), CHI '21 — Data Cascades in High-Stakes AI

SUTRA 6 · CHAPTER II

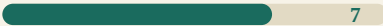
ECONOMY

The youth skilled in logic and learning are the true treasure of Bangladesh.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- AI/CS graduates per year (target trajectory)
- Brain-drain ratio of top-quintile STEM graduates
- R&D spending as % of GDP (0.3% → 1.0%)

PRIORITY ACTIONS

- Fund 1,000 Trust Scholarships tied to 5-year national service
- Embed AI ethics and computational thinking in secondary curricula
- Create returnee fellowships for diaspora AI researchers

Risk if ignored: The demographic dividend becomes a demographic liability: educated youth emigrate or remain underemployed.

Scholarly anchor: Sen (1999), Development as Freedom, OUP

SUTRA 7 · CHAPTER III

SECURITY

Where spies once walked in shadows, now algorithms walk in light unseen.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Existence of judicial warrant requirements for algorithmic surveillance
- Number of independent audits of state surveillance systems
- Public register of government AI surveillance deployments

PRIORITY ACTIONS

- Legislate judicial oversight for all algorithmic surveillance
- Create an independent Surveillance Review Board
- Publish an annual transparency report on state data requests

Risk if ignored: Unaccountable surveillance normalises; the state gains capabilities that outlive any government's good intentions.

Scholarly anchor: Zuboff (2015), J. of Information Technology 30(1)

SUTRA 8 · CHAPTER III

RESILIENCE

To foresee a flood — of water or of misinformation — is greater than to control it after.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Lead-time of AI flood forecasts (hours of warning)
- Mean time-to-detection of coordinated disinformation campaigns
- % of upazilas covered by early-warning dissemination

PRIORITY ACTIONS

- Deploy AI-enhanced flood forecasting nationally
- Stand up a Misinformation Early Warning Centre
- Integrate disinformation drills into disaster preparedness

Risk if ignored: Both literal and informational floods strike unwarned populations; response costs dwarf prevention costs.

Scholarly anchor: Vosoughi, Roy & Aral (2018), Science 359(6380)

SUTRA 9 · CHAPTER III

SECURITY

Let secrets travel in encryption, not in trust alone.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- % of inter-ministry traffic under end-to-end encryption
- Sovereign key-management infrastructure operational (Y/N)
- Annual penetration-test coverage of critical systems

PRIORITY ACTIONS

- Mandate end-to-end encryption for classified government traffic
- Build sovereign PKI and key-escrow infrastructure
- Quarterly red-team exercises on critical information systems

Risk if ignored: A single breach exposes decades of state communication; deterrence by secrecy collapses.

Scholarly anchor: Rid & Buchanan (2015), J. of Strategic Studies 38(1–2)

SUTRA 10 · CHAPTER IV

GOVERNANCE

The battlefield of the state is no longer land, but the minds of its people.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- National media-literacy index (baseline → target)
- % of school curricula including critical information literacy
- Penetration of verified public-information channels

PRIORITY ACTIONS

- National media and AI literacy campaign across all districts
- Critical information literacy in secondary education
- Strengthen public-service media as an epistemic anchor

Risk if ignored: *The population's information environment is shaped by foreign platforms and adversarial actors; cohesion erodes invisibly.*

Scholarly anchor: Bail et al. (2018), PNAS 115(37)

SUTRA 11 · CHAPTER IV

ETHICS

Divide not the people through careless algorithms, lest the state fracture without war.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Algorithmic-impact assessments filed per high-risk deployment
- Independent bias audits conducted per year
- Polarisation indices tracked in national surveys

PRIORITY ACTIONS

- Mandate Algorithmic Impact Assessments for public-facing systems
- Establish an independent algorithmic bias audit office
- Require platforms to publish recommender-system risk reports

Risk if ignored: *Engagement-optimised algorithms quietly amplify division until social fracture appears suddenly and irreversibly.*

Scholarly anchor: Angwin et al. (2016), ProPublica — Machine Bias

SUTRA 12 · CHAPTER IV

GOVERNANCE

Trust is the strongest embankment against unrest.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Public trust in AI-using institutions (annual survey)
- Complaint-resolution time for algorithmic decisions
- % of AI systems with published accountability documentation

PRIORITY ACTIONS

- Publish plain-language documentation for every public AI system
- Create an AI Ombudsman with binding remedy powers
- Institute participatory review panels with citizen members

Risk if ignored: *Each opaque failure compounds distrust; eventually even beneficial systems are rejected by a burned public.*

Scholarly anchor: Diakopoulos (2016), Communications of the ACM 59(2)

SUTRA 13 · CHAPTER V

ETHICS

Justice given by a machine must still be understood by a human.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- % of judicial AI outputs with human-readable reasoning
- Appeals filed against algorithm-assisted decisions and outcomes
- Human Rights Impact Assessments completed before deployment

PRIORITY ACTIONS

- Pilot judicial AI only with mandatory explanation interfaces
- Guarantee statutory right to contest algorithmic decisions
- HRIA required before any justice-sector AI deployment

Risk if ignored: *Defendants face verdicts no one can explain; due process becomes a formality and judicial legitimacy decays.*

Scholarly anchor: Dressel & Farid (2018), *Science Advances* 4(1)

SUTRA 14 · CHAPTER V

ETHICS

The State that sees all must not be feared by all.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Statutory limits on retention of citizen surveillance data
- Citizen access requests fulfilled within statutory window (%)
- Independent oversight body operational and reporting publicly

PRIORITY ACTIONS

- Codify purpose limitation and retention limits in law
- Give citizens access and correction rights over state-held data
- Annual public reporting by the oversight authority

Risk if ignored: *Chilled speech, performative conformity, and a brittle state that mistakes silence for stability.*

Scholarly anchor: UNESCO (2021), *Recommendation on the Ethics of AI*

SUTRA 15 · CHAPTER V

ETHICS

To control thought unjustly is a tyranny deeper than chains.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Regulation of psychologically-targeted political advertising
- Disclosure requirements for AI-personalised political content
- Detection rate of covert influence operations in elections

PRIORITY ACTIONS

- Ban undisclosed psychographic political targeting
- Mandate provenance labels on synthetic political media
- Empower the Election Commission with AI-forensics capacity

Risk if ignored: *Elections remain procedurally free while substantively manipulated; sovereignty of the voter quietly ends.*

Scholarly anchor: Matz et al. (2017), *PNAS* 114(48); Epstein & Robertson (2015), *PNAS* 112(33)

SUTRA 16 · CHAPTER VI

SECURITY

Neighbours are not only those across borders, but those across networks.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Mapped dependency index across cloud, model, and chip layers
- % of critical services with multi-vendor failover
- Bilateral digital-cooperation agreements in force

PRIORITY ACTIONS

- Commission a national digital dependency audit
- Adopt multi-vendor and exit-clause requirements in procurement
- Negotiate data-adequacy arrangements with key partners

Risk if ignored: Invisible dependencies become leverage in a crisis; a sanctions decision in a foreign capital can switch off domestic services.

Scholarly anchor: Dafoe (2018), AI Governance: A Research Agenda, Oxford

SUTRA 17 · CHAPTER VI

ECONOMY

Ally with those who share technology, not merely treaties.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Technology-transfer clauses in signed AI partnerships
- Joint research outputs with partner institutions per year
- Bangladeshi researchers in international AI consortia

PRIORITY ACTIONS

- Make capability transfer a precondition of AI MoUs
- Co-fund joint AI labs with ASEAN, EU, and Gulf partners
- Join international open-model and open-data alliances

Risk if ignored: Paper partnerships proliferate while capability stays abroad; Bangladesh signs much and learns little.

Scholarly anchor: Lee (2018), AI Superpowers, HMH

SUTRA 18 · CHAPTER VI

SECURITY

A company with data may wield more power than a state without it.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Effective tax and audit reach over platform revenues in BD
- Legal instruments compelling local platform accountability
- Market concentration of AI gateways serving Bangladeshi users

PRIORITY ACTIONS

- Enact platform accountability legislation with local representation rules
- Coordinate Global-South positions on platform governance
- Develop sovereign alternatives for critical public functions

Risk if ignored: Policy becomes negotiation with entities larger than the state, on terms the state cannot see or set.

Scholarly anchor: Zuboff (2019), The Age of Surveillance Capitalism

SUTRA 19 · CHAPTER VII

SECURITY

War is now silent — fought in circuits, currencies, and narratives.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- National cyber-incident mean time-to-respond
- Critical infrastructure entities meeting security baselines (%)
- Information-warfare doctrine adopted by armed forces (Y/N)

PRIORITY ACTIONS

- Publish a national cyber and information warfare doctrine
- Mandate security baselines for critical infrastructure operators
- Create a joint civil-military information defence command

Risk if ignored: The first decisive blow of a future conflict lands before anyone recognises a war has begun.

Scholarly anchor: Rid & Buchanan (2015), J. of Strategic Studies

SUTRA 20 · CHAPTER VII

SECURITY

Victory lies in shaping the enemy's choices, not destroying his armies.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Strategic-communications capacity assessments
- Scenario exercises on narrative conflict per year
- Counter-influence response time during crises

PRIORITY ACTIONS

- Build strategic communications capability in key ministries
- Run annual narrative-warfare scenario exercises
- Pre-position verified rapid-response information channels

Risk if ignored: Adversaries set the narrative frame in every crisis; the state is permanently reactive and disbelieved.

Scholarly anchor: Fang (2026), The Economist — machine audiences

SUTRA 21 · CHAPTER VII

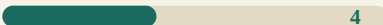
SECURITY

The best war is unseen and therefore undeniable.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Attribution capability: % of major incidents attributed within 90 days
- Forensics specialists per million population
- International attribution partnerships in force

PRIORITY ACTIONS

- Invest in national digital forensics and attribution labs
- Join collective-attribution coalitions with partner states
- Pursue UN norms criminalising covert AI information operations

Risk if ignored: Deniable aggression goes unanswered; impunity invites escalation against the least defended targets.

Scholarly anchor: Cavelti & Wenger (2020), Contemporary Security Policy 41(1)

SUTRA 22 · CHAPTER VIII

ECONOMY

Control the flow of goods, and you shape dependence; control the flow of data, and you shape destiny.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Cross-border data-flow agreements with reciprocity clauses
- % of national data value-chain captured domestically
- Localisation requirements for strategic data classes

PRIORITY ACTIONS

- Classify strategic data and set graduated localisation rules
- Negotiate reciprocal data-flow terms in trade deals
- Meter and value national data flows in official statistics

Risk if ignored: Bangladesh exports raw data and imports finished intelligence — the classic colonial trade pattern, digitised.

Scholarly anchor: Goldfarb & Tucker (2019), J. of Economic Literature 57(1)

SUTRA 23 · CHAPTER VIII

ECONOMY

Digital currency is not coin, but command.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- CBDC pilot design with privacy safeguards (status)
- Interoperability of national payments with regional rails
- % unbanked population reached by digital finance

PRIORITY ACTIONS

- Pilot a privacy-preserving retail CBDC under Bangladesh Bank
- Mandate open APIs and interoperability in digital payments
- Extend digital finance to rural women as a priority cohort

Risk if ignored: Monetary infrastructure is built by others, embedding foreign visibility and control into every transaction.

Scholarly anchor: Stiglitz (2002), American Economic Review 92(3)

SUTRA 24 · CHAPTER VIII

ECONOMY

Let others build upon your systems, and they shall rise within your influence.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Third-party services built on national digital public infrastructure
- Regional adoption of Bangladeshi-origin standards or stacks
- Open-source contributions from national projects

PRIORITY ACTIONS

- Build digital public infrastructure as open, extensible platforms
- Export the Bangla-LLM stack to Bangla-speaking communities globally
- Champion Global-South interoperability standards

Risk if ignored: Bangladesh remains a permanent tenant on others' platforms, paying rent in data, fees, and dependence.

Scholarly anchor: Rochet & Tirole (2003), J. of the European Economic Assoc. 1(4)

SUTRA 25 · CHAPTER IX

RESILIENCE

Floods will come — of water, of code, of chaos. Prepare before they rise.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- National AI-incident response plan tested per year
- Backup and continuity coverage of critical registries (%)
- Crisis-simulation exercises including AI failure modes

PRIORITY ACTIONS

- Adopt a National AI Incident Response Framework
- Test continuity of critical systems under simulated AI failure
- Cross-train disaster management and cyber response teams

Risk if ignored: *The first major AI-system failure cascades unchecked through unprepared institutions.*

Scholarly anchor: Scott (1998), Seeing Like a State, Yale UP

SUTRA 26 · CHAPTER IX

RESILIENCE

A resilient state has many layers, like the Bengal delta it stands upon.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Redundancy depth of critical digital services (layers)
- Local-capacity index: districts able to operate degraded-mode services
- Diversity of suppliers across the national AI stack

PRIORITY ACTIONS

- Architect critical services for graceful degradation
- Devolve operational capacity to division and district levels
- Enforce supplier diversity in strategic procurement

Risk if ignored: *A monolithic stack fails monolithically; one outage becomes a national standstill.*

Scholarly anchor: Walker et al. (2004), Ecology & Society

SUTRA 27 · CHAPTER IX

RESILIENCE

Speed of adaptation is greater than strength of position.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Policy iteration cycle-time (months from evidence to amendment)
- Civil-service reskilling throughput per year
- Adoption lag behind frontier practice (months)

PRIORITY ACTIONS

- Institutionalise rapid policy-learning loops with sunset clauses
- Fund continuous reskilling as a budget line, not a project
- Benchmark adoption lag annually against regional peers

Risk if ignored: *Static advantages evaporate; the state optimises yesterday's problems with growing precision.*

Scholarly anchor: Acemoglu & Restrepo (2019), J. of Economic Perspectives 33(2)

SUTRA 28 · CHAPTER X

GOVERNANCE

He who shapes the story of technology shapes the will of the people.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Reach of national AI public-engagement programmes
- Sentiment tracking of AI narratives in Bangla media
- Citizen participation in AI policy consultations

PRIORITY ACTIONS

- Run a national conversation on AI in Bangla, beyond elites
- Commission Bangla-language creative work on AI futures
- Mandate public consultation for major AI policies

Risk if ignored: *The AI narrative is written abroad; fatalism or hype displaces deliberate national choice.*

Scholarly anchor: Fang (2026), *The Economist*

SUTRA 29 · CHAPTER X

RESILIENCE

The next power will arise unseen — prepare before it is named.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Horizon-scanning reports delivered to Cabinet per year
- Foresight exercises covering frontier technologies
- Budget share allocated to anticipatory capability

PRIORITY ACTIONS

- Establish a National Foresight Office in the Cabinet Division
- Run annual frontier-technology scenario planning
- Maintain optionality: avoid irreversible single-vendor lock-ins

Risk if ignored: *The next disruption finds Bangladesh as unprepared as the last; catch-up costs compound.*

Scholarly anchor: Dafoe (2018), *AI Governance: A Research Agenda*

SUTRA 30 · CHAPTER X

ETHICS

Balance innovation with restraint, as the river balances flow with banks.

URGENCY



FEASIBILITY



IMPACT



KEY INDICATORS

- Time-to-permit for sandboxed AI ventures
- High-risk deployments blocked or remediated by review
- Innovation index vs. harm-incident index, tracked jointly

PRIORITY ACTIONS

- Pair every enabling measure with a proportionate safeguard
- Publish joint innovation-and-harm dashboards
- Review the regulatory perimeter annually with all stakeholders

Risk if ignored: *Either banks without flow (stagnation) or flow without banks (harm); both end in lost legitimacy.*

Scholarly anchor: Floridi et al. (2018), *Minds and Machines* 28(4)

Summary Matrix: The Thirty Sutras at a Glance

Colour intensity encodes score magnitude. Read down each column to identify national priorities; read across each row to characterise a single Sutra's strategic profile.

SUTRA	URGENCY	FEASIBILITY	IMPACT	THEME
S1	8	7	9	Governance
S2	9	6	9	Governance
S3	7	7	8	Governance
S4	10	6	10	Economy
S5	7	8	8	Economy
S6	9	7	10	Economy
S7	8	5	9	Security
S8	9	8	9	Resilience
S9	8	7	8	Security
S10	9	5	10	Governance
S11	8	5	9	Ethics
S12	8	6	10	Governance
S13	8	6	9	Ethics
S14	8	5	9	Ethics
S15	7	4	10	Ethics
S16	7	7	8	Security
S17	6	7	8	Economy
S18	8	5	9	Security
S19	9	5	9	Security
S20	6	5	8	Security
S21	7	4	8	Security
S22	8	6	9	Economy
S23	7	6	8	Economy
S24	6	6	8	Economy
S25	9	7	9	Resilience
S26	7	7	8	Resilience
S27	8	6	9	Resilience
S28	7	7	8	Governance
S29	6	5	9	Resilience
S30	8	6	9	Ethics

Triage reading. The highest-urgency, highest-impact cluster comprises Sutras 4 (data sovereignty), 6 (human capital), 10 (epistemic security), and 25 (incident preparedness) — these constitute the non-deferrable core of the national agenda. The high-feasibility quick wins are Sutras 5 (data annotation economy), 8 (early warning), and 28 (national narrative). The capability-constrained frontier — high impact, low current feasibility — comprises Sutras 15 (epistemic autonomy), 18 (platform power), and 21 (attribution), where international cooperation substitutes for capacities Bangladesh cannot yet build alone.